DEFENCE JOURNEY

NDA Mock Test 3

Practice test in the NDA pattern with answer key and explanations

NDA | MOCK 3

TIP: Time: 40 minutes. Questions: 30 (Maths 12, English 8, General Knowledge 10). Marking: +1 for a correct answer, −1/3 for a wrong answer, 0 if left blank. This is a short practice test in the NDA pattern, not the full-length paper. Attempt it without looking at the notes, then check the answer key on the last page.

Part A: Mathematics

Q1. If n(A) = 20, n(B) = 15 and n(A ∩ B) = 5, then n(A ∪ B) is

(A) 35
(B) 40
(C) 25
(D) 30

Q2. The modulus of the complex number 3 + 4i is

(A) 25
(B) 7
(C) 5
(D) 1

Q3. The sum to infinity of the GP 8, 4, 2, 1, … is

(A) 32
(B) 12
(C) 14
(D) 16

Q4. The value of ${}^{7}C_{3}$ is

(A) 35
(B) 42
(C) 70
(D) 21

Q5. If A is a 3 × 3 matrix with |A| = 2, then |3A| is

(A) 162
(B) 6
(C) 18
(D) 54

Q6. The value of tan 45° + cot 45° is

(A) 1
(B) 2
(C) √2
(D) 0

Q7. From a point 50 m from the foot of a tower, the angle of elevation of its top is 45°. The height of the tower is

(A) $50\sqrt{3}$ m
(B) 100 m
(C) 50 m
(D) 25 m

Q8. The equation of the line passing through (1, 2) and (3, 6) is

(A) $y = 3x$
(B) $y = x + 1$
(C) $y = x/2$
(D) $y = 2x$

Q9. The eccentricity of a parabola is

(A) 0
(B) 1
(C) 2
(D) 1/2

Q10. The value of $\lim_{x \to 0} (\sin 3x)/x$ is

(A) 1/3
(B) 0
(C) 1
(D) 3

Q11. The minimum value of $f(x) = x^2 - 4x + 5$ is

(A) 1
(B) 0
(C) 2
(D) 5

Q12. The vectors $a = 2i + j$ and $b = i - 2j$ are

(A) At 60°
(B) Parallel
(C) At 45°
(D) Perpendicular

Part B: English

Q13. Choose the word closest in meaning to CANDID.

(A) Rude
(B) Frank
(C) Shy
(D) Secretive

Q14. Choose the word opposite in meaning to TIMID.

(A) Shy
(B) Weak
(C) Fearful
(D) Bold

Q15. Fill in the blank: No sooner had he arrived ___ it began to rain.

(A) when
(B) then
(C) than
(D) that

Q16. What does the idiom 'to spill the beans' mean?

(A) To lose money
(B) To reveal a secret
(C) To cook well
(D) To waste food

Q17. One word for 'a person who does not believe in God':

(A) Theist
(B) Optimist
(C) Atheist
(D) Cynic

Q18. Fill in the blank: He is ___ honest man.

(A) no article
(B) the
(C) a
(D) an

Q19. Fill in the blank: If I ___ a bird, I would fly.

(A) am
(B) were
(C) will be
(D) was

Q20. Choose the correct passive form of: 'They are building a bridge.'

(A) A bridge is being built by them.
(B) A bridge has been built by them.
(C) A bridge was being built by them.
(D) A bridge is built by them.

Part C: General Knowledge

Q21. Who discovered penicillin?

(A) Edward Jenner
(B) Robert Koch
(C) Louis Pasteur
(D) Alexander Fleming

Q22. The Constitution of India came into force on

(A) 2 October 1950
(B) 26 January 1950
(C) 26 November 1949
(D) 15 August 1947

Q23. Which layer of the atmosphere contains the ozone layer?

(A) Exosphere
(B) Mesosphere
(C) Troposphere
(D) Stratosphere

Q24. Who is known as the 'Iron Man of India'?

(A) Subhas Chandra Bose
(B) Bal Gangadhar Tilak
(C) Sardar Vallabhbhai Patel
(D) Jawaharlal Nehru

Q25. The smallest bone in the human body is the

(A) Stapes
(B) Femur
(C) Tibia
(D) Radius

Q26. The largest planet in the solar system is

(A) Earth
(B) Saturn
(C) Jupiter
(D) Neptune

Q27. The Param Vir Chakra is awarded for

(A) Scientific research
(B) Highest gallantry in war
(C) Best sportsperson
(D) Peacetime bravery

Q28. The Battle of Plassey was fought in

(A) 1757
(B) 1526
(C) 1764
(D) 1857

Q29. The SI unit of electrical resistance is

(A) Volt
(B) Ampere
(C) Watt
(D) Ohm

Q30. Which gas is used to make soft drinks fizzy?

(A) Carbon dioxide
(B) Nitrogen
(C) Hydrogen
(D) Oxygen

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	2	C	3	D	4	A	5	D
6	B	7	C	8	D	9	B	10	D
11	A	12	D	13	B	14	D	15	C
16	B	17	C	18	D	19	B	20	A
21	D	22	B	23	D	24	C	25	A
26	C	27	B	28	A	29	D	30	A

Explanations

Q1. (D) $n(A \cup B) = 20 + 15 - 5 = 30$.

Q2. (C) $|z| = \sqrt{(3^2 + 4^2)} = 5$.

Q3. (D) $a = 8$, $r = 1/2$. $S = a/(1 - r) = 8/(1/2) = 16$.

Q4. (A) $7!/(3!\,4!) = (7 \times 6 \times 5)/6 = 35$.

Q5. (D) $|kA| = k^3|A|$ for a 3×3 matrix $= 27 \times 2 = 54$.

Q6. (B) $1 + 1 = 2$.

Q7. (C) $\tan 45° = h/50 = 1$, so $h = 50$ m.

Q8. (D) Slope $= (6 - 2)/(3 - 1) = 2$. Using (1, 2): $y - 2 = 2(x - 1)$, so $y = 2x$.

Q9. (B) For a parabola, the distance from the focus equals the distance from the directrix, so $e = 1$.

Q10. (D) $(\sin 3x)/x = 3 \times (\sin 3x)/(3x) \to 3 \times 1 = 3$.

Q11. (A) $f(x) = (x - 2)^2 + 1$. The minimum is 1 at $x = 2$.

Q12. (D) $a \cdot b = 2 \times 1 + 1 \times (-2) = 0$, so they are perpendicular.

Q13. (B) A candid person is honest and open.

Q14. (D) Timid means lacking courage. Bold means brave.

Q15. (C) *No sooner* is followed by *than*. *Hardly/scarcely* is followed by *when*.

Q16. (B) It means to let out a secret by accident or on purpose.

Q17. (C) An atheist does not believe in God. A theist believes in God.

Q18. (D) *Honest* begins with a vowel sound (the 'h' is silent), so 'an' is used.

Q19. (B) For an unreal or imaginary condition, *were* is used with all subjects.

Q20. (A) Present continuous active becomes is/are + being + past participle.

Q21. (D) Alexander Fleming discovered penicillin in 1928.

Q22. (B) It was adopted on 26 November 1949 and came into force on 26 January 1950.

Q23. (D) The ozone layer lies in the stratosphere, 15 to 35 km above the Earth.

Q24. (C) Sardar Patel integrated the princely states after Independence.

Q25. (A) The stapes in the middle ear is the smallest bone.

Q26. (C) Jupiter is the largest planet.

Q27. (B) It is India's highest wartime gallantry award. The Ashoka Chakra is the highest in peacetime.

Q28. (A) Robert Clive defeated Siraj-ud-Daulah at Plassey in 1757.

Q29. (D) Resistance is measured in ohms (Ω).

Q30. (A) Carbon dioxide dissolved under pressure gives the fizz.